

Working of an EV

The three main components of an EV are: electric motor, controller and battery. The electric motor needs no oil, no tune-ups and, since there are no tailpipe emissions, it does not necessitate any smog checks.

When you switch on the vehicle, current is passed from the battery. The controller takes power from the battery and delivers it to the motor. Before passing the current to the motor, the controller converts 300V DC into a maximum of 240V AC, 2-phase power, which is suitable for the motor.

The accelerator pedal hooks to a pair of potentiometers (variable resistors). These potentiometers provide the signal that tells the controller how much power it is supposed to deliver.

When you push on the accelerator pedal, a cable from the pedal connects to these two potentiometers. There are two potentiometers for safety. The controller reads both potentiometers and makes sure that their signals are equal. If not, then it does not operate. This arrangement guards against a situation where a potentiometer fails in the full-on position.

The controller can deliver zero power (when the car is stopped), full power (when the driver floors the accelerator pedal) or any power level in between. The electric motor then converts electrical energy into mechanical energy, which moves the vehicle forward.

Other components of an EV are electric engine, regenerative braking and drive system.

Battery. The battery is the key component of an EV because of the following:

- Range of the vehicle depends almost entirely on HV battery.
- It is the heaviest electrical component.
- It is also the most expensive electrical component.

A battery electric vehicle (BEV) uses a battery that can be charged using ordinary grid electricity at a specialised power station. Besides conventional lithium-ion battery technologies, there are lead-acid, nickel-metal-hydride (NiMH) and zebra.

Lithium-ion. Today's EVs mostly

use lithium-ion batteries—the same technology that can be found in a laptop or e-book reader. These batteries are now considered to be the standard for modern BEVs. There are many types of lithium-ion batteries and each has different characteristics. But vehicle manufacturers are focused on variants that have excellent longevity.

Compared to other mature battery technologies, lithium-ion offers many benefits. For example, it has

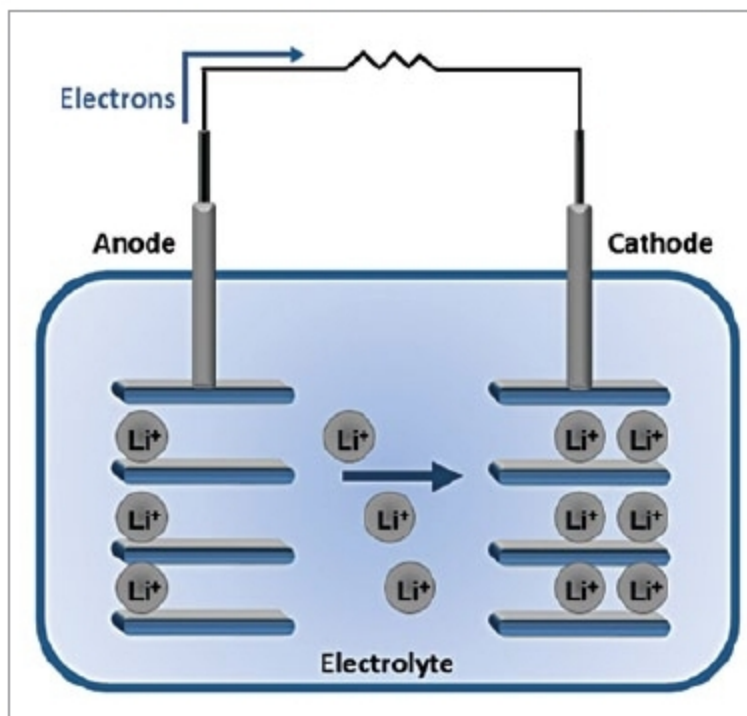


Fig. 2: Lithium-ion battery powering a device

excellent specific energy (140Wh/kg) and energy density, making it ideal for BEVs. It is also excellent in retaining energy with a self-discharge rate (five per cent per month), which is an order of magnitude lower than NiMH.

However, lithium-ion batteries also have some drawbacks. Six significant problems with current lead-acid battery technology are:

- These are heavy; a typical lead-acid battery pack weighs 463.6kg (1000 pounds) or more.
- These are bulky; the car used as an example here has 50 lead-acid batteries, each measuring roughly 15.24x20.32x15.24cm (6x8x6-inch).
- These have a limited capacity; a typical lead-acid battery pack might hold 12 to 15 kilowatt-hours of electricity, giving the car a range of only 80.5km (50 miles) or so.
- These are slow to charge; typical recharge times for a lead-acid pack range between four to ten hours

for full charge, depending on battery technology and charger.

- These have a short life—three to four years, perhaps 200 full charge/discharge cycles.

- These are expensive—perhaps US\$ 2000 for a typical battery pack.

When you place the battery in a device, positively-charged lithium-ions are attracted to and move towards the cathode. Once it is bombarded with these ions, the cathode becomes more positively-charged than the anode, and this attracts negatively-charged electrons.

As electrons start moving towards the cathode, these are forced to go through the device and use the energy of the electrons flowing towards the cathode to generate power. You can think of this kind of like a water wheel, except instead of water flowing, electrons are flowing.

Lithium-ion batteries are great because these are rechargeable. When the battery is connected to a charger, lithium ions move in the opposite direction

as before. As these move from the cathode to the anode, the battery is restored for another use.

Lead-acid and NiMH batteries. Both lead-acid and NiMH batteries are mature battery technologies. These were originally used in early EVs such as General Motor's EV1. However, these are now considered to be obsolete with regards to their use as the main source of energy storage in BEVs. Lead-acid batteries have been used in conventional petroleum-driven vehicles and are relatively inexpensive. However, these have a poor specific energy of 34Wh/kg.

NiMH batteries are considered to be superior as these can have up to double the specific energy—68Wh/kg—compared to lead-acid batteries. This allows EVs that utilise NiMH batteries to be significantly lighter, leading to reduced energy cost for propelling the BEVs. NiMH batteries also have greater energy density compared to lead-acid batteries, which allows